

# Using taplo To Lint And Format TOML Files

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TOML stands for “Tom’s Obvious Minimal Language”.

From [TOML’s website](#):

TOML aims to be a minimal configuration file format that’s easy to read due to obvious semantics. TOML is designed to map unambiguously to a hash table. TOML should be easy to parse into data structures in a wide variety of languages.

Many configuration files are `.toml` files (e.g. `pyproject.toml` or `air.toml`) and I have engaged with `.toml` files for several years.

Within the past month, I learned about `taplo` (<https://taplo.tamasfe.dev/>) which is a “...versatile, feature-rich TOML toolkit” written in Rust (<https://github.com/tamasfe/taplo>) that can be used for linting and formatting `.toml` files. One can add this tool as a `pre-commit` hook with a dedicated configuration file, e.g.

(in `taplo.toml`)

```
[formatting]
align_entries = true
column_width  = 79

# see https://taplo.tamasfe.dev/configuration/formatter-options.html
# for more formatting options
```

(in `.pre-commit-config.yaml`)

```
- repo: https://github.com/ComPWA/taplo-pre-commit
  rev: v0.9.3
  hooks:
    - id: taplo-format
    - id: taplo-lint
      args: ["--no-schema"]
```

The options available to format are given in the table below:

| option                | description                                                                                                                    | default value     |
|-----------------------|--------------------------------------------------------------------------------------------------------------------------------|-------------------|
| align_entries         | Align entries vertically. Entries that have table headers, comments, or blank lines between them are not aligned.              | false             |
| align_comments        | Align consecutive comments after entries and items vertically. This applies to comments that are after entries or array items. | true              |
| array_trailing_comma  | Put trailing commas for multiline arrays.                                                                                      | true              |
| array_auto_expand     | Automatically expand arrays to multiple lines when they exceed <code>column_width</code> characters.                           | true              |
| array_auto_collapse   | Automatically collapse arrays if they fit in one line.                                                                         | true              |
| compact_arrays        | Omit whitespace padding inside single-line arrays.                                                                             | true              |
| compact_inline_tables | Omit whitespace padding inside inline tables.                                                                                  | false             |
| inline_table_expand   | Expand values (e.g. arrays) inside inline tables.                                                                              | true              |
| compact_entries       | Omit whitespace around <code>=</code> .                                                                                        | false             |
| column_width          | Target maximum column width after which arrays are expanded into new lines.                                                    | 80                |
| indent_tables         | Indent subtables if they come in order.                                                                                        | false             |
| indent_entries        | Indent entries under tables.                                                                                                   | false             |
| indent_string         | Indentation to use, should be tabs or spaces but technically could be anything.                                                | 2 spaces<br>(" ") |
| trailing_newline      | Add trailing newline to the source.                                                                                            | true              |
| reorder_keys          | Alphabetically reorder keys that are not separated by blank lines.                                                             | false             |
| reorder_arrays        | Alphabetically reorder array values that are not separated by blank lines.                                                     | false             |
| reorder_inline_tables | Alphabetically reorder inline tables.                                                                                          | false             |
| allowed_blank_lines   | The maximum amount of consecutive blank lines allowed.                                                                         | 2                 |
| crlf                  | Use CRLF line endings.                                                                                                         | false             |

Find below an example `.toml` file (the `pyproject.toml` file for my home site) before and after linting and formatting (using the aforementioned `taplo.toml` and `.pre-commit-config.yaml` settings):

Before Taplo

```

[project]
name = "o957-github-io"
version = "0.0.1"
authors = [
    {name = "0957", email = "127630341+0957@users.noreply.github.com"},
]
description = "My home page, which contains some information about my
↪ person, activities, and work."
license = "Apache-2.0"
readme = "README.md"
keywords = ["personal", "website", "data-science", "paleontology"]
requires-python = ">=3.11"
dependencies = [
    "blackjax>=1.6.2",
    "jax>=0.10.2",
    "numpyro>=0.21.0",
]

[project.urls]
Repository = "https://github.com/0957/0957.github.io"
Issues = "https://github.com/0957/0957.github.io/issues"
"Author GitHub" = "https://github.com/0957"

[tool.ruff.lint.mccabe]
max-complexity = 12

[tool.vulture]
paths = ["scripts"]

```

After Taplo

```

[project]
name = "o957-github-io"
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